

ULIASTAI, China

WMO: 509150

Lat: 45.5126N

Lon: 116.9630E

Elev: 841

StdP: 91.62

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.8	-28.6	-34.7	0.2	-30.1	-32.6	0.2	-28.1	9.6	-11.4	8.2	-12.8	1.0	20	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	33.1	17.3	31.1	16.7	29.2	16.3	20.2	27.1	19.2	26.2	18.3	25.3	3.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	14.4	22.7	17.1	13.5	21.8	16.0	12.6	21.2	62.1	27.1	58.4	26.3	55.6	25.1	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	7.9	6.9	-33.4	36.2	2.4	2.5	-35.1	37.9	-36.5	39.4	-37.9	40.8	-39.7	42.6		
(5)			-33.5	21.5	2.4	1.4	-35.2	22.6	-36.6	23.4	-38.0	24.2	-39.7	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-19.7	-14.7	-5.6	4.9	12.6	19.0	22.2	20.0	13.3	3.6	-8.0	-16.9		
	DBStd	15.38	4.74	5.70	6.14	5.39	4.91	4.02	3.15	3.49	4.56	5.22	6.52	4.77		
	HDD10.0	3894	922	693	483	169	27	1	0	0	19	206	541	835		
	HDD18.3	6002	1180	926	741	403	189	39	3	20	161	456	791	1093		
	CDD10.0	1210	0	0	1	16	108	271	379	311	117	8	0	0		
	CDD18.3	275	0	0	0	0	11	59	124	72	9	0	0	0		
	CDH23.3	3052	0	0	0	14	213	703	1217	762	141	2	0	0		
	CDH26.7	1102	0	0	0	2	69	259	478	263	31	1	0	0		
(14) Wind	WSAvg	2.8	2.3	2.4	3.1	3.6	3.6	2.9	2.6	2.5	2.6	2.8	2.6	2.4		
(15) Precipitation	PrecAvg	249	2	2	4	8	19	40	71	59	29	11	3	3		
	PrecMax	417	7	7	11	47	68	97	170	192	76	44	13	9		
	PrecMin	153	0	0	0	0	0	10	21	16	1	0	0	0		
	PrecStd	63	2	2	3	9	16	21	33	39	15	9	3	2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.8	4.0	15.4	25.4	32.0	34.8	36.5	35.4	29.8	22.0	11.4	0.1		
		MCWB	-6.7	-1.1	4.9	9.8	14.1	16.1	18.7	18.7	14.7	9.9	3.4	-3.0		
	2%	DB	-7.0	0.8	11.1	21.2	28.4	31.9	33.5	31.7	27.2	18.4	7.2	-5.0		
		MCWB	-9.2	-2.8	2.9	8.3	12.3	15.7	18.5	17.3	13.9	8.5	1.2	-7.0		
	5%	DB	-9.6	-2.3	7.6	18.1	25.4	29.6	31.3	29.4	24.7	15.3	4.3	-7.4		
		MCWB	-11.0	-5.2	0.9	6.9	11.2	15.5	17.9	16.7	12.9	6.9	-0.3	-9.1		
	10%	DB	-11.9	-5.4	4.4	15.2	22.6	27.5	29.3	27.5	22.3	12.7	1.5	-9.6		
		MCWB	-13.0	-7.6	-0.9	5.5	10.3	14.8	17.6	16.2	12.1	5.5	-2.1	-10.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.5	-0.4	5.5	11.0	15.2	18.7	21.9	21.1	17.7	11.0	3.9	-3.3		
		MCDB	-3.9	3.3	13.9	23.1	26.6	27.7	28.5	30.4	24.8	20.4	10.3	0.2		
	2%	WB	-9.2	-2.9	3.1	8.8	13.6	17.7	20.7	19.6	15.6	9.1	1.6	-6.9		
		MCDB	-7.2	0.6	10.8	19.8	24.6	26.4	27.3	26.3	23.2	16.8	6.6	-5.0		
	5%	WB	-11.1	-5.3	1.0	7.3	12.3	16.9	19.7	18.7	14.3	7.5	-0.2	-9.0		
		MCDB	-9.6	-2.3	7.2	16.9	22.9	24.9	26.7	25.0	21.9	14.6	4.2	-7.5		
	10%	WB	-13.0	-7.6	-0.8	5.8	11.0	16.1	19.0	17.8	13.1	5.9	-2.0	-10.9		
		MCDB	-11.9	-5.4	4.3	14.9	21.3	23.8	26.3	24.3	20.5	12.3	1.4	-9.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	12.6	12.9	14.0	14.1	12.3	11.3	12.1	13.5	12.7	11.1	9.9		
		MCDBR	12.6	15.5	17.1	18.5	18.6	15.6	14.6	14.9	16.6	16.7	13.9	12.0		
	5% WB	MCWBR	11.0	12.2	10.7	9.7	7.8	5.3	4.3	4.9	7.2	9.1	9.4	10.3		
		MCWBR	12.6	15.5	16.4	17.1	15.5	11.9	11.3	11.0	13.6	15.5	13.6	11.9		
(40) Clear-Sky Solar Irradiance	taub		0.264	0.291	0.342	0.402	0.475	0.493	0.444	0.413	0.364	0.332	0.292	0.271		
	taud		2.287	2.231	2.150	2.036	1.882	1.951	2.165	2.217	2.283	2.281	2.276	2.247		
	Ebn at Noon		846	885	880	852	799	785	822	834	848	822	796	787		
	Edh at Noon		79	102	129	159	192	181	144	132	113	97	80	74		
(44) All-Sky Solar Radiation	RadAvg		2.12	3.29	4.64	5.76	6.37	6.28	6.16	5.78	4.73	3.37	2.22	1.69		
	RadStd		0.14	0.20	0.22	0.30	0.40	0.51	0.33	0.23	0.27	0.21	0.15	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	-1.54	N/A	-0.51	-0.93	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	-1.82	N/A	N/A	-0.65	N/A	N/A	N/A	N/A

Nomenclature: See separate page